

Joshua,

I looked over the page on github and the primary issue I see with this project is its lack of accessibility. University policy and executive direction require that all digital content added to the website (or any other platform under the NMU name) meet the WCAG 2.1 AA standards for digital accessibility. I didn't run a full audit of this page but from a cursory review, I see mobile display issues, contrast issues, and keyboard functionality issues. Beyond those functional issues, the data represented by the charts and graphs is not machine readable. The one instance I found where some of the data from the charts is represented in the code there are no aria labels present describing each element's purpose. This means that anyone trying to interact with this page using assistive technology, like a screen reader, will be unable to do so. Because of these issues, embedding this content in its current state using an iframe is something I cannot approve.

If we were to embed these charts and graphs as images, the contrast issues would need to be resolved and text-based descriptions of each item would need to be provided. Assuming we could provide accessible alternatives in the form of images with descriptions, in theory we could then provide a link to the github page where users could see the interactive dashboard. We would want to provide an accessibility disclaimer with that link but as long as we're providing this accessible alternative we should be legally compliant. I'm not an expert on how this law will be interpreted and enforced so anything that we would do like this we would probably want to run by risk management

I realize this is probably not what you were hoping to hear. Digital accessibility has become an important topic over the past few years and upcoming federal regulations have required us to become much more conscious of what we're putting on the website and how our users interact with it. We've had to adjust or re-think many projects because accessibility was not considered early on in development.

-Eric

Then I wrote

Eric,

First, thank you so much for the detailed response. Second, would consider taking another look at the site in a week, so I can address the important accessibility issues you raised?

There is a set of circumstances that preclude the option to embed images. That is. I am leaving NMU, unfortunately. This is a last gift that I have been working on, both for NMU and for the U.P. Thus, the automation that uses API keys to collect weekly data from the Bureau of Economic Analysis and the Bureau of Labor Statistics and visualize them in a dashboard is necessary, as I won't be here to update static images. In the worst viable case, we could point to an external site with a disclaimer. Would that work?

Also, is there a website nationally and then one at NMU that describes the national and university level accessibility requirements, respectively, in detail? My goal, should you be willing to reconsider, would be to spend the holiday weekend making the site compliant, and, though this could be a work in progress, I'd like to do it once.

Thanks,

Joshua

Then he wrote

This is the WCAG 2.1 guidelines: <https://www.w3.org/TR/WCAG21/>. Within those guidelines there are three levels so success criterion, A, AA, and AAA. Our content has to be up to the AA standards.

This sites provides a more practical explanation of the WCAG standards and is what I generally use when looking for a quick overview of the various requirements: <https://aaardvarkaccessibility.com/wcag-plain-english/>

Finally, this Educause article does the best job of summarizing the upcoming regulations, the limited exceptions and their likely impact on higher ed: <https://er.educause.edu/articles/2024/6/web-and-mobile-app-accessibility-regulations>.

In regards to your question about providing a link and a disclaimer, I don't believe that will be acceptable under this regulation or executive direction. If that were a possible solution, institutions would likely use it for many projects instead of investing the resources to address the accessibility issue. The primary issue that I see with this project is that there is no way to interact or understand the data without use of visual elements on these pages. Since these elements are all being created by an API using a database on the back end, maybe it would be possible to make accessible alternatives in the form of data tables used to build the visual elements. This guide provides guidance on creating accessible graphs and charts: <https://informationaccessgroup.com/making-graphs-and-charts-more-accessible/>.

I hope this information helps. I don't know if it will be enough to help fix this project but these are the best resources I can provide.

-Eric